**4.** **ROD AND BEAD (8 points)** — *Eero Ristolainen, Jaan Kalda.* A bead is threaded on a rigid frictionless rod. The rod rotates with constant angular velocity $\omega$ about a horizontal axis that is perpendicular to the rod and passes through a fixed point on it Gravitational acceleration is $g$.

**i)** *(1 point)* With suitable initial conditions, the trajectory of the bead is a circle. Sketch this trajectory in the plane perpendicular to the rotation axis, in a coordinate system with the axis at the origin.

**ii)** *(1 point)* Given that the trajectory is exactly a circle, find its radius.

**iii)** *(2 points)* The circular trajectory of the previous tasks is unstable. To stabilise it, one attaches a spring that produces a restoring force $-kx$ when the bead is displaced by $x$ along the rod from the axis of rotation. For what values of the stiffness $k$ can the bead move along a *stable* circular trajectory, and what is the radius of such a trajectory?

**iv)** *(4 points)* Next, the spring is removed, the bead is given a charge $q$, and a uniform electric field of strength $E$ whose vector rotates with angular velocity $\Omega$ in the same direction as the rod is introduced. With suitable initial conditions, the bead follows the trajectory shown below. Find $\Omega$, $g$, and the ratio $a_E = Eq/m$, where $m$ is the mass of the bead, in terms of $\omega$ and $d$, the side length of the grid squares shown in the figure. You may take measurements from the figure.

